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THIRD SEMESTER B. TECH EXAMINATION, DECEMBER - 2024

INTRODUCTION TO AEROSPACE VEHICLES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5 x2=10**
1.	**a.**	What do you mean by a perfect gas?	L2	CO1	**02**
	b.	Outline the difference between Dihedral and Anhedral.	L2	CO2	**02**
	c.	State the use of Collective pitch in helicopters.	L2	CO3	**02**
	d.	Define center of pressure & Center of gravity.	L2	CO4	**02**
	e.	List the types of Planet Orbits and Earth orbits used for payload transfer.	L2	CO5	**02**
		PART - B			**6x5=30**
2.	**a.**	Discuss the contribution of Sir George Cayley to the field of aviation.	L2	CO1	**05**
		OR			
	b.	Assume that, at a point on the wing of the Concorde supersonic transport aircraft, the air temperature is -10°C and the pressure is 1.7×10^4 N/m^2. Calculate the density at this point.	L3	CO1	**05**
3.	**a.**	What do you mean by Airfoil; state its characteristics?	L2	CO2	**05**
		OR			
	b.	What do you mean by Stability; list the types of stability?	L2	CO2	**05**
4.	**a.**	Differentiate between Angle of Attack and Angle of Incidence.	L2	CO2	**05**
		OR			
	b.	What is meant by V –n diagram and state its importance.	L2	CO2	**05**
5.	**a.**	Explain in detail about the Helicopter flight controls with a neat sketch.	L2	CO3	**05**
		OR			
	b.	What is meant by Autorotation and state the need of it in helicopters?	L2	CO3	**05**
6.	**a.**	List the secondary & auxiliary flight controls and describe their functions.	L2	CO4	**05**
		OR			
	b.	Explain in detail about the FADEC with a neat sketch.	L2	CO4	**05**
7.	**a.**	What is a Missile and how do we classify missiles?	L2	CO5	**05**
		OR			
	b.	Differentiate between Tactical and Strategic Missiles.	L2	CO5	**05**

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		PART - C			6x10=60
8.	**a.**	What is a Gyroscope? Explain the Gyroscopic instruments used in aircraft with a neat sketch.	L2	CO1	**10**
		OR			
	b.	Derive the standard atmospheric equations for pressure, density at Isothermal layer.	L2	CO1	**10**
9.	**a.**	On a rectangular fluid element, a force P is applied on the bottom face and this force travels through a volume where (Base and Sides are unity and Height is dhg) the force exerted at the top face is P+dp. Derive the equation for h using this case study.	L3	CO1	**10**
		OR			
	b.	What is meant by a Sweep back and variable sweep aircraft? What are the advantages and disadvantages of this aircraft design?	L2	CO2	**10**
10.	**a.**	Illustrate the parts of an airplane and discuss in detail their primary functions.	L2	CO2	**10**
		OR			
	b.	Illustrate and explain Fly by light and Fly by wire system with a neat sketch.	L2	CO4	**10**
11.	**a.**	Explain the parts of a helicopter with its primary functions with a neat sketch.	L2	CO3	**10**
		OR			
	b.	Explain Reaction Torque and Describe various Anti Torque mechanisms.	L2	CO3	**10**
12.	**a.**	Aircraft manufacturer has designed an aircraft with high wing configuration but they have a problem with fuel feed system. Suggest a fuel feed system for this aircraft and explain it with a neat sketch.	L3	CO4	**10**
		OR			
	b.	Explain about the operating principle of Basic Hydraulic System and its components.	L2	CO4	**10**
13.	**a.**	Due to foggy weather heading indicator has failed and there is only less visibility for the pilot, kindly suggest an alternate system to the pilot to find the heading and land the aircraft.	L3	CO4	**10**
		OR			
	b.	Classify rockets based on propellants and explain the operating principle of Hybrid Propellant rocket with its advantages and disadvantages.	L2	CO5	**10**